BioSpecInfo

Functional Specifications — BioSpecInfo

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Functional Specifications — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v231`
Purpose	Describe what the product does, for whom, under which rules and with which limits.

1. Product description

BioSpecInfo is an **interactive cheminformatics** platform for university study of chemistry, biochemistry, pharmacology and astrochemistry.

The whole scientific logic — interpretation of molecular structures, spectral prediction, calculations, 2D and 3D visualisation — runs **in the student's browser**. There is no server: the application installs as an app, works with no network, and transmits no data.

1.1 Objectives

Objective	How it is pursued
Total accessibility	No registration, no cost, no server. Whoever opens the link is already using it.
Independence from the network	Everything needed is bundled and cached: it works in a lecture hall, on a train, in a library with no signal.
Verifiable correctness	Scientific data are put through automatic checks; the limits of the models are declared rather than hidden.
Privacy by construction	There is no backend that could collect data: the absence of collection is not a promise, it is a consequence of the architecture.

1.2 What the product is not

Not a medical device	It does not support diagnosis, prescription or dosing (Reg. EU 2017/745)
Not a certified analytical instrument	The spectra are predictions, not measurements
Not a collaborative platform	No accounts, no sharing, no synchronisation between devices

2. Actors

Actor	Description	What they can do
Student	Primary user. Studying for a university exam.	Every function: consultation, calculation, quizzes, spaced repetition, AI assistant

Actor	Description	What they can do
Teacher	Uses the app as teaching support	The same functions; uploading their own material into the File Manager
Technical assessor	A company or committee examining the work	Reading the documentation and re-running the verification benches

There are no technical roles: there is no authentication, so there is no distinction of permissions. The File Manager has a local protection that is a **deterrent**, not an access control (see §6).

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3. Functional areas

The application has **120 sections**, grouped by area of study.

3.1 Spectroscopy

Function	Description
Spectroscopy centre	A molecule is entered (SMILES, name, formula) and one obtains IR, $^{1}H/^{13}C$ NMR, MS, UV-Vis, Raman, 2D and 3D structure, functional groups
Visual IR spectra	Characteristic bands by class of compound
Mass spectrum	Fragmentation patterns and diagnostic rules
UV-Vis	Chromophores, λmax, Woodward's rules
Per-atom NMR prediction	Draw or type a structure and get ^{1}H and ^{13}C with WHICH atoms produce each signal; interrogable spectrum (zoom, intensity, click a peak), assignment table, SMILES/InChI history, CSV/JCAMP-DX/PNG exports
Two-dimensional NMR	COSY, edited HSQC and HMBC built by walking the graph; clicking a spot highlights the atoms producing it. It is not a simulation of the experiment, and the panel says so
Open any file, and the program decides	One entry point, with no extension filter: open what you have and the program **states what it decided the file is**. A document (PDF, photograph, Word, OpenDocument, PowerPoint, Excel, text) opens, **everything** it contains is shown (every page drawn, all the text), the recognised spectroscopic data appear in a table **with their provenance**, and the working proceeds step by step; a JCAMP-DX file or two columns of numbers become a measured spectrum; an image does both. Whatever could not be read is declared
Header language switcher	A round button top right, at the title's height, with the flag of the language in use; it opens into a disc with the fourteen flags on two shells and, at the centre, the language name and **how much it is translated, measured**. The actual switch stays in the languages module: this is only the way to call it

Function	Description
Spectrum from a figure (digitisation)	From a photograph of a chart recorder, a crop from a paper or the instrument screen, the trace is extracted from the **pixels**: the axis frame is recognised and only its inside is digitised, the line position is the **darkness-weighted centroid** (sub-pixel), the grid is worked around by continuity and darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10}T$. The chart, the peaks and the assignments appear by themselves and are redone at every change of the calibration; a diagnostics line says what was decided and why. CSV and JCAMP-DX outputs. Measured deviation: mean 2.2 cm^{-1}, worst 7.3 over seven constructed figures
3D model from the drawn structure	Coordinates built from the graph, carrying the predictor's own indices: a clicked peak lights up the atom in three dimensions too
⚛️ **Physics: eight computed sections**	**Schrödinger laboratory** (stationary states of wells, oscillators, Morse and double well; wave packet and tunnelling with Crank-Nicolson; vibrations of HCl, H_2, CO, N_2 in cm^{-1} compared with experiment); **live Hartree-Fock** (RHF/STO-3G in the browser, eight molecules, orbitals drawn, dissociation curves); **rovibrational spectroscopy** (P and R branches with temperature and isotopes, Raman with spin statistics, bond length from the lines); **statistical thermodynamics** (partition functions, S° and Cp against NIST-JANAF, populations); live Lennard-Jones **molecular dynamics**; **mathematics for chemistry** (FID → spectrum by FFT, Hückel, kinetics by Runge-Kutta, Arrhenius least squares); **crystallography** (14 Bravais lattices, Miller indices, powder pattern with absences); **radiochemistry** (Bateman, dating, Weizsäcker). Every number is checked by `test_fisica` against the literature
📐 **Mathematics: eight sections on a symbolic engine**	A computer-algebra engine written for the app (`bsi-mat-cas.js`: exact fractions, canonical form, rule-based derivatives, antiderivatives verified by differentiating them, Taylor series by truncated power-series arithmetic, limits) underpinning: **basics** (significant figures and measurement rules, logarithms and pH, equations from linear to polynomial and the fractional ones of equilibrium, practice with marking); **function analysis** (domain, zeros, extrema, inflections, asymptotes, tangent; families and transformations); **derivatives, integrals and series** (rules used, Riemann, adaptive Simpson, Taylor, left and right limits); **linear algebra** in exact fractions with Gauss-Jordan steps, eigenvalues, plane transformations and reaction balancing as a null space; **differential equations** (direction field, order of methods, phase plane with equilibria classified from the Jacobian); **several variables** (3D surfaces, critical points, Maxwell relations, exact differentials); **complex numbers and Fourier**; **laboratory statistics** (t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD/LOQ)
🎓 **UniBA exams and advanced physical chemistry**	**Mathematics exam**: Istituzioni di Matematiche I and II syllabi explained, step-by-step solver for 16 exercise types, 16 solved papers (64 exercises), timed simulation. **Physics exam**: General Physics I and II syllabi, 48 standard problems solved with the user's data (laws, substitutions, units), 12 solved papers. **Electrodes and transition state** (Butler-Volmer, Tafel, Eyring) and **photochemistry, surfaces and polymers** (Jablonski, Stern-Volmer, Langmuir/Freundlich/BET, Mn/Mw, Carothers). **Chemistry exam** (🔬 category): 40 standard problems on four levels — basic (General and Inorganic), intermediate (Analytical), advanced (Physical Chemistry), expert (Sackur-Tetrode, Debye-Hückel, Pourbaix) — and 7 solved papers. Mathematics and Physics also gain an **advanced and expert** level (eigenvalues, ODE systems, Fourier, Laplace, triple integrals; driven oscillations, waves, Doppler, photoelectric, Compton, relativity, blackbody, particle in a box, tunnelling). The traces are built on the syllabi, they are not the official papers

Function	Description
🧭 **From high school to PhD**	A map of chemistry, physics and mathematics on four levels (high school, bachelor's, master's, PhD) with links to the tools. New solvers: **astrophysics** (10 models), **nuclear physics** (9), **quantum physics** (11, with variational and perturbation methods), **organic chemistry, spectroscopy and biochemistry** (10), **PhD** (Marcus, isotope effect, Wigner; residues, heat equation)
⚗️ ⚛️ 📐 **The three macro-areas**	At the head of the Chemistry, Physics and Mathematics categories. Each has an **assistant** to which you write the problem in natural language (English too): it recognises the kind of exercise or the model, reads numbers with units, formulas, reactions and lists, says what it understood and where each datum comes from, also solves the **inverse problem** and shows the working; the **catalogue** of every tool of the subject with search; the **levels** from high school to PhD. Spectra has the same solvers as tools
NOE and 2D NMR (tables)	Reference values for spatial correlations
Reference tables	Tabulated values for quick consultation
Spectrum quiz	Recognising the functional group from a spectrum

3.2 General and physical chemistry

Interactive periodic table · chemical calculator · reaction balancing · degrees of unsaturation · distribution diagrams · VSEPR · molecular orbitals · advanced thermodynamics · electrochemistry (Nernst) · pKa and buffers · laboratory calculator · constants and formulae.

3.2-bis Separation — chromatography

Chromatography section: a chromatogram computed from the parameters (theoretical plates, retention factor, selectivity, dead time) with an overview and an expanded view of the peak pair; a **van Deemter** curve with the three terms separated and the minimum computed in closed form; a **Kováts retention index** calculator; a table of the quantities with their working ranges; GC/HPLC comparison; a guide to detectors; diagnostics for peak-shape defects.

> **Rule applied.** The chromatogram is not an illustration: the Gaussians have σ = t_R/√N and the resolution shown is the one that would be measured on those peaks. The section explicitly declares what it does **not** do — predict a molecule's retention time from its structure, which would require experimental parameters of the stationary phase that cannot be derived from the SMILES alone.

3.2-ter Calibration and uncertainty

Calibration section: ordinary least-squares regression on data pasted from a spreadsheet, with slope and intercept **accompanied by their standard deviation**; a plot of the line with a 95 % confidence band and a **residual plot**; the unknown's concentration with a full

confidence interval (the three sources 1/M, 1/N and distance from the centroid are shown separately); **LOD and LOQ** per ICH Q2(R1); uncertainty propagation indicating which contribution dominates; significant-figure rules.

Two diagnostics that an r^2 does not provide:

Diagnostic	Method	Why r^2 is not enough
Non-linearity	F-test between line and parabola on the same data	An r^2 of 0.984 accompanies an obvious saturation curve: r^2 measures how much variance the line explains, not whether the line is the right model
Outlier	**Deleted** residual: the judging line is rebuilt without the point being judged	The naive criterion `

> **The second row is a trap met while writing the section**, not a textbook case: the 3 σ criterion was already written, and the example meant to demonstrate it was not being flagged. The check passed without seeing anything.

3.2-quater Cheminformatics and QSAR

Cheminformatics section: a workbench that goes from the raw set to a verdict on the model, entirely in the browser. SMILES are pasted in — with a name and an activity, where there are any — and six panels return what a cheminformatics group produces in a day.

Panel	What it does
Descriptors	Standardisation (largest fragment, canonical SMILES, deduplication), 43 RDKit descriptors, Lipinski's rule of five, Veber filter, QED recomputed — MinimalLib does not expose it
Similarity and groups	Morgan/ECFP4, RDKit, pattern and MACCS fingerprints; Tanimoto and Dice; similarity matrix; nearest neighbours; **Butina clustering** and **MaxMin** selection for diversity
Chemical space	PCA on the standardised descriptors, with explained variance per component and the **loadings** that say which descriptor moves which axis; **Bemis–Murcko** scaffolds with the molecule count per scaffold
QSAR model	Kernel ridge (Tanimoto kernel), kNN over Tanimoto or logistic regression; **scaffold split**; **grouped cross-validation** over 3, 5 or 10 folds; **model comparison on identical folds** with the trivial reference at the top; hyperparameters chosen by **nested validation**; **conformal prediction intervals**; **learning curve**; applicability domain
Search	Substructure search across the whole set with RDKit's `SubstructLibrary`, SMARTS or SMILES query, occurrence counts, eight ready queries
SAR table	R-group decomposition around a core: one row per molecule, one column per substitution position, with each substituent's median activity. Molecules without the core are **excluded**, not shown with empty cells
Matched pairs	Pairs differing by a **single** substitution, grouped by transformation with median, range and **number of concordant pairs**
Activity cliffs	Structurally close pairs with distant activities, ranked by **SALI**: these are the pairs every model gets wrong, and it is honest to show them

Panel	What it does
Structural alerts	PAINS and Brenk, with the matched fragment highlighted and the reason it is flagged

Three choices that separate a workbench from a demonstration:

Choice	Method	Why
Scaffold split	The test set contains **only scaffolds never seen** in training	A random split puts close analogues on both sides: the model recognises rather than predicts, and the R^2 that comes out does not survive the first new molecule
Null model by scrambling	The same model is retrained on **shuffled labels**, eight times; the verdict is the comparison	An R^2 of 0.4 can arise by chance when the molecules are few. If the random twins reach the same score, the model has learnt nothing — and the panel **says so**, instead of showing only the good number
Deduplication on the canonical form	`OC(=O)C` and `CC(=O)O` are the same molecule	The same molecule in training and in test is an information leak, and the written string is not enough to see it

> **The verdict is written out in words, not left to the reader.** The QSAR panel does not stop at the metrics: it states whether the model beat all its random twins, and by how much. Measured on the 28-inhibitor example: R^2 0.861 against a maximum of 0.574 among the eight twins — margin 0.287. On the same molecules with random labels the model does **not** beat the twins, and the panel says so in the same way.

The results can be taken away

Two exports, both formed in memory by the browser: there is no network request, which is why the note beside the buttons can say that no data leaves the device without asking anyone to take it on trust.

Export	Content
CSV table	One row per molecule: name, canonical and original SMILES, activity, scaffold, nine descriptors, QED, Lipinski violations, Veber outcome, alerts found. With cross-validation active, **predicted** and **residual** are added — and only then, because cross-validation is the one scheme that gives every molecule a prediction from a model that had not seen it
SDF	The structures with **generated 2D coordinates**, the name on each block's first line and the data fields attached: it is the format in which sets are exchanged between groups and programs. Verified by round-trip
Method report (.md)	What is needed to **redo** the same analysis: RDKit version, fingerprint, split type, number of molecules and scaffolds, **the seed**, the metrics, the null-model verdict, the folds one by one, and the declared limits

> **Why the seed is in the report.** Without it, "random split" and "scrambled labels" mean nothing: two runs would give different numbers and nobody could say which was the right one. The generator is a seeded mulberry32, and the same seed gives the same folds.

A measurement no other bench could make

`verifica-accessibilita` walks the 120 sections and measures what is **visible**. The six panels of this section live inside a container that stays `display:none` until the analysis has been run: that bench never saw them, and its "contrast 0" was not speaking about them.

`test_cheminfo` runs the analysis for real, opens the six panels in turn and measures WCAG contrast on every piece of text that appears there — **460 elements, 0 defects**. The formulas are rewritten inside the bench rather than taken from the application: a bench that asked the code under test how good its own contrast was would be measuring nothing.

The engine lives in `bsi-cheminfo.js`, is exposed as `window.BSIChem` and is verified by the `test_cheminfo` bench (189 checks) against hand-computable values.

3.3 Organic chemistry

Retrosynthesis (98 exercises) · synthesis and mechanisms with electron arrows · animated mechanisms · structure editor · *Organic Chemistry 1+2* module with 25 modules.

3.4 Biochemistry

Amino acids · biomolecules · animated metabolic pathways · enzyme kinetics (Michaelis-Menten) · 3D macromolecules · interactive biological structures · biochemistry guide · exam biochemistry.

The **biochemistry guide** is a document of its own (`Biochimica_Guida_Definitiva.html` , 8.2 MB) loaded into a frame by the section: 12 chapters plus the index, 78 subsections, 98 embedded figures, and two chapters devoted to the exam — "Oral exam guide" and "Real exam questions and correct answers". It contains **no scripts and makes no network requests**: it cannot raise JavaScript errors, and the alternative text of the 98 figures comes from the captions the author wrote, not from invented descriptions (see `docs/06` §3-ter).

The document closes with a **self-test of 12 multiple-choice questions**, recovered from the previous guide — where it was a JavaScript quiz — and rewritten **without a line of code**: the four options are in plain sight, the solution sits in a native `<details>` that opens on a click. In print the `<details>` disappears and a dedicated copy of the solution appears, so the printed copy does not lose the explanations.

3.5 Pharmacology

Function	Description
3D Atlas	One chooses a body region, a disease and a drug: the model highlights the target organ
Database	**263 drugs** with structure, molecular weight, mechanism, indications, adverse effects, class. For **36** the entry carries the ChEMBL identifier and the record's formula, and the formula is rebuilt from the graph and compared
Interactions	Checking interactions between active ingredients
Pharmacokinetics	Calculator for half-life, clearance, volume of distribution

Function	Description
Clinical cases	Diagnosis-therapy paths for teaching purposes

3.6 Astrochemistry

Molecules of the interstellar medium · exoplanets and JWST data · nebulae · stellar spectra · nucleosynthesis · cometary chemistry · quizzes.

3.6-bis Spectrum reader

You load a file from your own instrument, or paste a text, and it is read.

Function	Description
Formats	**JCAMP-DX** (`.jdx` , `.dx`) with full **ASDF** compression — PMAI for digits, DIF for differences, DUP for repeats — or two columns of numbers separated by spaces, commas or tabs
Integrity check	The JCAMP format prescribes that, after a run of differences, the next value written in full must equal the computed one. That check is performed here
Peak finding	By **prominence**, with noise estimated from the median absolute deviation (MAD × 1.4826) rather than the standard deviation, which one tall peak inflates
IR assignment	For each band, **all** compatible assignments among 24 ranges are listed, not the first
Mass spectra	A peak list is recognised as such; differences between ions are matched against 17 **neutral losses**
NMR	Integration of areas by chemical-shift region
Drawing	IR with **decreasing** wavenumbers, the way it is read

What it does not do, stated in the section itself: it does not deduce the structure. A set of bands is compatible with many molecules, and a program that produced a name from three peaks would make a claim the data cannot support.

3.6-ter Languages and chemical languages

Function	Description
Interface language	**14 languages** with a selector and a search field: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese, Arabic. Each one's coverage is **measured**, not declared
Writing direction	Arabic carries `dir="rtl"` ; every other language `ltr` , including on the way back
Chemical languages	**25 outputs** from one SMILES: formula and mass composition, canonical/flat/explicit-hydrogen SMILES, CXSMILES, InChI and InChIKey, the key's skeleton block, Murcko scaffold, CIP stereo tags, SMARTS and CXSMARTS, molfile V2000 (also aromatic) and V3000, XYZ, PDB, CML, JSON, and the **six fingerprints** (Morgan, MACCS, RDKit, atom pairs, topological torsions, pattern)
IUPAC naming	Over a **declared class** (acyclic, neutral, C H O N F Cl Br I, one kind of principal group, linear substituents). Outside that class it **refuses** instead of guessing

3.7 Study tools

Function	Description
Spaced repetition	SM-2 algorithm on cards generated from the contents
Quizzes	By subject, with score and history
Study path	A guided sequence of topics
Notes	Global and per section
Pomodoro	Study timer
File Manager	Personal archive of photos, documents and HTML pages

3.8 The "Spectra" assistant

An AI agent with **38 tools** that can interrogate the application itself: compute descriptors, generate spectra, look up constants, balance reactions, open sections. It supports ten providers, and stays usable when one of them fails (§5.2).

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4. Main flows

4.1 Analysing a molecule

```
1. Open the Spectroscopy centre
2. Enter the molecule:
     · SMILES                → used directly
     · name or formula       → resolution via PubChem (needs network)
     · drawing in the editor → SMILES generated
3. The application shows: 2D structure, functional groups, descriptors,
   predicted spectra (IR, NMR, MS, UV-Vis), drug-likeness
4. The 3D structure loads if there is a network (conformer from PubChem)
```

With no network: point 3 stays complete for SMILES entered directly; resolution by name and the 3D structure are unavailable, and the application says so instead of hanging.

4.2 Asking the assistant

```
1. Tap the "Spectra" button
2. On first use: choose a provider and enter an API key,
   or connect a proxy (and then no key is needed)
3. Type the question
4. The assistant answers, using its tools when needed:
   every tool used is shown, not hidden
```

If the provider does not answer: the assistant notes it, demotes it in the selection for 24 hours and moves to another configured provider. The 🔌 **Test** button measures in a few seconds which providers the device actually reaches, **without needing any key**.

4.3 Studying with spaced repetition

```
1. Open the study section
2. Generate a deck from the contents
3. For each card, rate how well it was remembered
4. The SM-2 algorithm computes when to show it again
5. On reopening, only the cards that are due appear
```

4.4 Archiving your own material

```
1. File Manager → local password
2. Upload photos, documents or HTML pages
3. The material stays ON THE DEVICE (IndexedDB), it is not transmitted
```

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5. Business rules

5.1 Scientific data

Rule	Rationale
A molecular structure enters the database only if the recomputed weight agrees with the literature one (±0.6 u)	A plausible but wrong datum is indistinguishable from a correct one for the user
If a structure does not pass verification, the entry stays **without a structure**	A wrong datum replaced by another wrong one is no progress
Every absence must be **recorded with its reason**	An unjustified exception becomes a shortcut
Spectra are deterministic	Two spectra of the same molecule must be comparable
An ambiguous physical constant is not resolved at random	A wrong value given with confidence is worse than a request for clarification

5.2 AI assistant

Rule	Behaviour
A provider that does not answer is remembered for **24 hours**	Then it is retried: today's failure is not a sentence
A retired model is learned from the error message	Hand-coded lists grow stale
An imposed wait longer than **10 seconds** switches provider, if a ready one exists	Beyond that threshold the user believes the app has frozen
One turn at a time	Two overlapping submissions corrupt the history
Paid providers are not offered as free ones	Without active billing one hits the limit at the first question

5.3 User data

Rule	Behaviour
At most 30 conversations, 100 messages each	Beyond that, `localStorage` fills up and **every** subsequent save fails
Images up to 20 MB	Beyond that, explicit refusal; the others proceed
Selective deletion by group	Wiping everything to free space is an avoidable loss
A failed write **must** be declared	"✅ Saved" when it is not true is worse than an error

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6. Current limitations

Declared, not hidden. The detail is in `09-Release-Conformance-Statement.md` §4.

#	Limitation	Practical consequence
1	**2 entries with no structure to show** (it was 6)	Ivermectin is a mixture of B1a/B1b homologues and Coartem a combination of two active ingredients: a single SMILES notation does not represent them, and for these two the 2D/3D and predicted spectra are missing. Digoxin, vincristine and tacrolimus (topical and systemic) now have their structure, taken from ChEMBL and verified against the molecular weight
2	**Predicted, not measured, spectra**	Useful for recognising functional groups; not for formal identification
3	**No synchronisation**	Data do not pass from one device to another, and there are no copies on a server
4	**File Manager: deterrent, not security**	On a static site, whoever reads the source bypasses any page-side control. The source carries only the SHA-256 digest of the password, never the password
5	**Functions that need a network**	IUPAC name, CAS, GHS, 3D conformers, AI assistant
6	**Accessibility automated, not complete**	Contrast, accessible names, labels, heading hierarchy across 13 pages and all 120 sections: **0 defects**, including text over gradients (judged against the worst stop). What stays outside is text inside SVGs and text over a real background image, counted on every run, and everything requiring human judgement
7	**Chromatography: no retention prediction**	The section computes resolution, efficiency and indices **given** k, α and N; it does not predict k from a structure, which would require experimental parameters of the stationary phase
8	**Partial code coverage**	Measured: **49.89 %** of statements over the widest path a bench walks. It is not whole-battery coverage, and it is statements, not branches; see `08-Traceability-Matrix.md` §8
9	**Verification on Chromium only**	Firefox and WebKit are tested by hand, not by a bench

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7. Possible developments

Not commitments: directions consistent with the architecture.

Area	Proposal	Why
Data	The 2 remaining entries have no structure to show	It is not a gap to close: ivermectin is a mixture, Coartem a combination. At most one could show the **components**, declared as such
Spectra	Comparison with experimental databases (SDBS/NIST) inside the app	It would make the distance between prediction and measurement visible
Accessibility	Add `axe-core` alongside, for the rules the bench does not implement	ARIA roles, tab order and focus management stay outside the current measurement
Export	Full export of the user's data	The only possible remedy for the absence of backups
Assistant	A preconfigured proxy for those who do not want to publish one	It would remove the last barrier to AI access

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Document updated to version `bsi-v231`.